For all projects, students must use a medium to large dataset, containing at least 200,000 records (rows).

The datasets listed below are provided only as examples or suggestions. Students are free to select other datasets, provided that they satisfy the project requirements regarding data size and structure.

If one of the suggested datasets does not contain a sufficient amount of data, it should not be used, and an alternative dataset should be selected.

Project 1: Large-Scale Recommendation Systems

The goal of this project is to design a data-driven recommendation system capable of suggesting relevant items to users based on historical interaction data. The system should analyse large, structured datasets describing user behaviour (ratings, purchases, listening events, etc.) and identify patterns that can support personalized recommendations.

Recommendation systems are widely used in modern digital platforms such as Netflix, Amazon, and Spotify, where they help users discover relevant content among thousands or millions of available items.

This project explores how large behavioural datasets can be processed and analysed to support recommendation strategies, representing a practical application of intelligent techniques for large, structured data.

Students may choose one of the following datasets or propose a similar one:

- MovieLens Dataset – movie ratings and metadata
  https://grouplens.org/datasets/movielens/
- Amazon Product Reviews Dataset – product ratings and reviews
  https://nijianmo.github.io/amazon/index.html
- Spotify Tracks Dataset – music metadata and popularity features
  https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset
- Online Retail Dataset – customer transactions from an e-commerce store
  https://archive.ics.uci.edu/ml/datasets/online+retail

Project 2: Image Analysis and Visual Pattern Recognition

The objective of this project is to develop a system capable of analysing large collections of images to identify patterns, categories, or visual similarities.

Image analysis plays an important role in many applications such as medical imaging, autonomous systems, visual search engines, and product recognition in e-commerce platforms.

Large image datasets can be analysed using computer vision techniques to automatically recognize objects, classify images, or detect patterns. By processing visual data at scale, intelligent systems can support tasks such as object detection, image classification, or visual similarity search.

This project investigates how large image datasets can be processed and analysed using intelligent techniques for visual data understanding.

Students may use datasets such as:

- CIFAR-10 Image Dataset
  https://www.cs.toronto.edu/~kriz/cifar.html
- Fashion-MNIST Dataset
  https://github.com/zalandoresearch/fashion-mnist
- Chest X-Ray Images Dataset
  https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia

Project 3: Large-Scale Customer Segmentation and Behavioural Analysis

The goal of this project is to analyse large datasets describing customer behaviour in order to identify groups of users with similar characteristics or purchasing patterns.

Customer segmentation is widely used in marketing, e-commerce, finance, and digital platforms to better understand user behaviour and design personalized services.

Companies collect large amounts of structured data describing customer activity, including purchases, browsing behavior, and demographic information. By analyzing these datasets, it is possible to identify patterns and group customers with similar behaviors.

This project explores how large structured datasets can be used to discover behavioral segments and hidden patterns in customer data.

Students may use datasets such as:

- Online Retail Dataset
  https://archive.ics.uci.edu/ml/datasets/online+retail
- Instacart Market Basket Analysis Dataset
  https://www.kaggle.com/datasets/psparks/instacart-market-basket-analysis
- E-commerce Customer Behavior Dataset
  https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store

Project 4: Large-Scale Text Toxicity Detection

The objective of this project is to develop a system capable of detecting toxic or harmful language in large collections of online text. The system should analyse large textual datasets and identify different types of toxic content such as insults, threats, hate speech, or abusive language.

Toxicity detection is widely used in social media platforms, online communities, and discussion forums to automatically moderate content and maintain healthy digital environments.

Online platforms generate massive volumes of user-generated text, including comments, posts, and discussions. Among these interactions, some messages may contain toxic or harmful language that negatively affects online communities.

By analysing large text datasets, it is possible to develop models that automatically detect toxic content and classify different types of abusive behavior. Such systems are used by platforms such as YouTube, Reddit, Twitter, and online gaming communities to support content moderation.

This project explores how large textual datasets can be processed and analyzed to automatically detect toxic language, representing a practical application of text mining and natural language processing techniques.

Students may use datasets such as:

- Jigsaw Toxic Comment Classification Dataset
  https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge
- Wikipedia Toxic Comments Dataset
  https://www.kaggle.com/datasets/julian3833/jigsaw-toxic-comment-classification-challenge
- Hate Speech and Offensive Language Dataset
  https://www.kaggle.com/datasets/mrmorj/hate-speech-and-offensive-language-dataset

Project 5: Fraud Detection in Financial Transactions

The objective of this project is to develop a system capable of detecting fraudulent financial transactions by analysing large datasets of payment records. Fraud detection systems are widely used by banks, payment platforms, and financial institutions to identify suspicious activity and prevent financial losses.

Financial systems process millions of transactions every day, making manual fraud detection impossible. By analyzing transaction patterns, it is possible to identify anomalies that may indicate fraudulent behavior.

This project explores how large transaction datasets can be analyzed using intelligent techniques to identify suspicious patterns and detect fraud. Such systems are critical components of modern digital banking and payment infrastructures.

Students may use datasets such as:

- Credit Card Fraud Detection Dataset
  https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud
- IEEE Fraud Detection Dataset
  https://www.kaggle.com/c/ieee-fraud-detection
- Synthetic Financial Dataset for Fraud Detection
  https://www.kaggle.com/datasets/ealaxi/paysim1

Project 6: Large-Scale Anomaly Detection in System Logs

The objective of this project is to develop a system capable of detecting anomalies or unusual patterns in large datasets. Anomaly detection systems are widely used in areas such as cybersecurity, financial monitoring, industrial systems, and infrastructure management, where identifying abnormal behavior is critical.

Large digital systems continuously generate massive volumes of data describing operational activity. Within these datasets, certain records may represent rare or abnormal events, such as system failures, unusual user behavior, or security incidents.

This project investigates how machine learning techniques can be used to identify anomalies in large structured datasets, allowing automatic detection of unusual patterns or outliers.

Examples of datasets:

- HDFS Log Dataset
  https://github.com/logpai/loghub
- BGL System Log Dataset
  https://github.com/logpai/loghub
- Thunderbird System Log Dataset
  https://github.com/logpai/loghub

Project 7: Predictive Modelling for Customer Churn

The goal of this project is to develop a system capable of predicting customer churn, meaning the likelihood that a customer will stop using a service or product. Churn prediction is widely used in industries such as telecommunications, subscription platforms, banking, and digital services.

Companies collect large amounts of data describing customer behavior, including service usage, transactions, and account activity. By analyzing these datasets, it is possible to identify patterns associated with customers who are likely to leave a service.

Machine learning models can be trained to predict churn based on historical behavior, allowing companies to intervene and improve customer retention strategies.

This project explores how large customer datasets can be used to build predictive models for churn detection.

Possible datasets:

- Telco Customer Churn Dataset
  https://www.kaggle.com/datasets/blastchar/telco-customer-churn
- IBM Customer Churn Dataset
  https://www.kaggle.com/datasets/shubh0799/churn-modelling
- E-commerce Customer Behavior Dataset
  https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store

Project 8: Trip Duration Prediction for Ride-Sharing Platforms

The goal of this project is to develop a system capable of predicting the duration of taxi or ride-sharing trips based on historical trip data.

Accurate trip duration estimation is important for ride-sharing services because it helps improve route planning, pricing strategies, and user experience.

Transportation datasets typically include information such as pickup location, drop-off location, timestamps, and trip distance. By analysing these attributes, machine learning models can estimate the expected duration of a trip under similar conditions.

This project explores how large transportation datasets can be used to build predictive models for travel time estimation.

Possible datasets:

- NYC Taxi Trip Duration Dataset
  https://www.kaggle.com/c/nyc-taxi-trip-duration
- NYC Taxi Trips Dataset
  https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Project 9: Real-Time Event Stream Analysis

The objective of this project is to develop a system capable of processing and analysing event streams generated by online platforms in near real-time. Such systems are widely used in modern data infrastructures to monitor user activity, detect patterns, and generate analytics from continuous data streams.

Many modern applications generate continuous streams of data, such as user clicks, application logs, or transaction events. Instead of processing data in large batches, streaming systems analyse data as it arrives, enabling real-time insights.

This project explores how streaming data can be ingested, processed, and analyzed to extract useful patterns and metrics, similar to systems used by large-scale platforms such as Uber, Netflix, or Amazon.

Students may use datasets such as:

- Clickstream Data for Web Analytics
  https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store
- Online Retail Dataset
  https://archive.ics.uci.edu/ml/datasets/online+retail

Project 10: Online Trend Detection from Social Media Data

The objective of this project is to analyse large volumes of social media data in order to detect emerging topics or trends over time.

Trend detection is widely used in news analytics, marketing analysis, political communication, and social media monitoring.

Social media platforms generate massive amounts of user-generated content, including posts, comments, and hashtags. By analysing this data over time, it is possible to detect emerging topics, viral content, and changes in public attention.

This project explores how large textual datasets can be analysed to identify trending topics and temporal patterns in online discussions.

Possible datasets:

- Twitter Sentiment / Tweet Dataset
  https://www.kaggle.com/datasets/kazanova/sentiment140
- Reddit Comments Dataset
  https://www.kaggle.com/datasets/reddit/reddit-comments-may-2015

Project 11: Duplicate Ads Detection in Online Marketplace

Online marketplaces allow users to easily buy and sell items. One of the largest platforms, Avito, hosts millions of listings where sellers advertise products. To gain more visibility, some sellers post the same advertisement multiple times with slightly different descriptions or photos. These duplicate ads make it harder for buyers to find genuine listings.

The objective of this project is to develop a machine learning model that can automatically detect duplicate advertisements on the platform.

The goal is to identify listings that represent the same item even if their text descriptions or images are slightly different. Improving duplicate ad detection will help reduce clutter in search results and make it easier for buyers to find legitimate listings.

Dataset: Avito Duplicate Ads Detection | Kaggle

Project 12: Analysis of Freelance Job Market

Analyse and model patterns within a large-scale freelance marketplace dataset with more than 1 million entries. The goal of this project is to find relevant information about freelance job market by choosing to do one or more of the following:

- contract price prediction
- job success probability prediction (e.g. a freelancer winning a contract)
- identify trends in skill demand across time
- job description clustering
- freelance job recommendation system (e.g. design a system to recommend a suitable job for a certain freelancer based on historical data)
- automatic skill extraction (e.g. extract skill information that freelancers have based on their description)

Dataset: https://www.kaggle.com/datasets/asaniczka/freelance-contracts-dataset-1-3-million-entries